Repairing Dysfunction in Active Complex Systems:
Therapy in medicine, high technology and society

Miro R to David G
On January 9, 2020

Complex Systems

We have some ideas about the emergence of order such as life and its molecular constituents. But, we have only vague ideas how complex systems, such as human organism, or society, work. What algorithm assures functional homeostasis (harmony) by regulating interactions among molecular and cellular constituents that run life, and what is the algorithm of interactions among persons and institutions that run human society and grow diverse cultures? If we knew such algorithms, we could more effectively avoid and heal diseases and create a more harmonious civilisation. With such knowledge, public health and world peace could flourish.

However, we do know from biology and engineering that the maintenance and growth of complex systems require repair, replacement and removal of system's constituents that are wrong, dysfunctional, or toxic to the system. This is so-called turn-over of systems' constituents that maintains their high quality.

The challenge is how to go about repairing dysfunctions of complex systems without crashing the whole system (like, if one had to repair or change aeroplane engine in flight)? Fearing system's crash hinders the initiatives to repair or replace dysfunctional societal institutions, or add new ones, for a more harmonious and efficient societal functionality.
It is wise to be cautious, but cautiousness that prevents repair of dysfunction, or implementation of the necessary change, can also lead to the crash of the whole system (organism's death or collapse of civilisation).
We need better science of complex biological systems - a better understanding of homeostasis (harmony) of catalytic activities and molecular interactions in functional cells, of cells in functional organisms, and of people and institutions in a functional society. Otherwise, costly radical system's resetting may follow in the form of societal crises, devastating wars or destructive revolutions.

At the level of human organism, system's complexity - and our insufficient knowledge about it - is revealed by unexpected secondary effects of drugs intended to mitigate system's dysfunction (i.e., disease). Modern drugs are targeted to particular molecular species, or cellular structures, suspected to be the cause of dysfunction. To no surprise for a systems biologist, "side-effects" or "secondary effects" arise signalling that we do not know enough about human systems biology to cure diseases.

Diagnosing and repairing societal dysfunctions is immensely important but poorly developed area of research. This is so because there is no tradition of scientifically planned political experimentation on the small scale ("pilot experiments""). Instead, there is direct action at the level of entire countries based either on beliefs, or on historical experience, rather than on experimental knowledge in actual societal space and real time. Let us concentrate here on systems aspects of human biology in relation to medicine.

How Healthy are the Medicine and Biomedical Research?

Poor patients have no alternative but to be patient. However, biomedical scientists should not be patient because it is unethical to be complacent with dismal success in curing prevalent deadly diseases and keep doing more of the same unproductive research. The reason to be disappointed about the results is that, for over half century, biomedical research received the lion's part of global research funds - specifically for its mission to mitigate prevalent diseases. One could question how productive (healthy?) are biomedical research and medicine themselves (as diagnosed from patients' viewpoint) by a rigorous evaluation of the efficacy of new drugs and current medical practices, and of the productivity of the mainstream biomedical research in providing knowledge necessary to mitigate age-related diseases.

After over half a century of monumental investments of human and financial resources into biomedical research - unequalled in the history of science - one can argue that the success in mitigating non-communicative diseases is, at best, mediocre. Age-related diseases remain incurable. In my opinion, such failure is due to poor thinking (resulting in wrong, or missing, concepts and inadequate scientific approaches) more than to the shortage of cutting-edge technologies. Arguably, shortage of concepts and common sense thinking lead mainstream biomedical scientists to an unproductive path of analytical studies of already advanced diseases.

Disease is the consequence - not the cause - of biological dysfunction. So, why study the consequences? Studying advanced diseases is too late and too complicated to be useful for health: it is a mere accountancy, book-keeping, of countless molecular changes and perturbations derived from the initial cause that triggered latent disease a couple of decades before its onset. The sterility of fashionable detailed molecular "mechanistic studies" of diseases can be compared with the uselessness to understanding and avoiding avalanches by identifying countless damages done by avalanches. Hence, studying diseases is disease-relevant, but health-irrelevant.

The return to common sense

To learn how to stay healthy by mitigating, i.e., avoiding, attenuating and reverting diseases (back to health), it would be reasonable to study what we want (health) rather than what we

don’t want (disease). There can be little doubt that knowing the biology of a single super-centenarian - a person evidently resistant to all of over one hundred of age-related degenerative and malignant diseases and conditions - would teach more about mitigation of age-related diseases than all world-wide studies of millions of patients put together.
While waiting for such health-relevant knowledge (by studying resilient healthy people), how do we go - now - about mitigating incurable diseases? Can we repair dysfunctional molecular and cellular systems or attenuate the effect of dysfunction?

Biological function and dysfunction reside in proteins

Gene alterations cause diseases only when they affect biological function. Vast majority of emerging random mutations in our genome do not cause diseases. They are either inconsequential silent mutations or evolution driving mutations - a process that created all life, including human.
Disease is caused by biological dysfunction, and proteins carry all biological functions, either directly or indirectly (via the synthesis of all bioactive molecules, short of metal cofactors, catalysts and electrolytes that act via their interaction with proteins). Hence, proteins that are defective or hyperactive (for whatever reason) are the predominant cause of biological dysfunction and of age-related disease. Therefore, proteins should be the priority targets in disease prevention and healing. But, which proteins and by what kind of drugs?

Good news for eventual success of novel drugs is that, normally, protein molecules must be renewed (turned-over) by synthesis of new, and breakdown of old, proteins. Namely, proteins are chemically relatively unstable and therefore functionally short-lived. Conveniently, damaged dysfunctional cellular proteins are preferentially turned over. Thus, if old dysfunctional proteins were systematically eliminated (by proteasome and autophagy) and newly produced proteins protected from the cause of their dysfunction, then the protein-based protein-turnover system itself would improve and the entire proteome rejuvenated and functionally maintained without the necessity to know how the whole system works. This is an interesting prediction.

When the cause of protein dysfunction is of chemical nature, usually oxidative damage, the maintenance of function should work by the prevention of oxidative damage during at least one round of protein turn-over. When protein dysfunction is due to a gene mutation, then either such gene mutation, or its consequence upon protein structure and function, should be repaired or phenotypically neutralized (“buffered”, i.e., attenuated or compensated), respectively. By far the most desirable and least risky way of mitigating age-related diseases is their avoidance by active, targeted, prevention (as below).

An alternative to the mainstream approach

Mainstream cell biology and biomedical research look for any molecular changes in diseased animals and humans, or rather in cells of their diseased tissues. The most frequent molecular parameters measured in diseased cells are: the amount and quality of particular proteins, metabolites, emerging mutations, telomere length variation, aberrant physiological protein or DNA modifications (e.g., methylation, phosphorylation, acetylation, etc.), protein misfolding and aggregation, or altered protein-protein interactions. By now, thousands of molecular changes in diseased tissues have been identified and some have been chosen as drug targets. The results remain disappointing.

An alternative approach is based on the following premises:

(i) The chemistry of aging is oxidative damage to proteins leading to progressive loss of function, dysfunction or to the acquisition of toxic, cell-killing, function.

(ii) Mutations that predispose to age-related diseases are phenotypically silent, or latent, usually for a larger fraction of lifetime than is the period of their phenotypic expression (from disease onset to death).

(iii) Hence, preventing phenotypic expression, i.e., the pathological consequences, of mutations is health-wise as good as eliminating mutation, but should be safer and easier to achieve.

(iv) When an inborn mutation eliminates a disease-relevant protein, or its function, the baby is born sick; we speak of a syndrome. When mutation does not directly damage protein function but causes protein fragility towards oxidative damage, the mutation is called “silent mutation”. As such, it will shorten the functional lifetime of the protein and cause disease onset after decades of life; it will be a predisposition to age-related disease.

At this point, in the light of four premises, finding small chemical “correctors” of protein structure, called molecular chaperones becomes a priority. Such chemical chaperones taken systemically, can spread through the organism and protect misfolded proteins from oxidation, or fix the correct protein structure - in spite of mutations! This approach looks more feasible and less risky than correcting mutations in the DNA*. The latter approach requires that all, or most, of organism’s cells - or just cells of the organ that produces the dysfunctional protein - be corrected.
The strategy of attenuation of the effect of mutation applies only to the cases of “silent” mutations (or polymorphisms) in proteins, but not to mutations (or epigenetic gene silencing) that eliminate or severely damage the protein. In this latter case, there is no physical entity from which the biological function can be rescued.

Fortunately, most or all mutations that predispose to age-related diseases are of the first kind (i.e., protein polymorphisms phenotypically silent in youth) and are therefore subject to

disease attenuation/mitigation by protection of the fragile mutant protein from oxidative damage, or by buffering the molecular defect by chaperoning.

The advantage of the latter approach is that it can be implemented now, since it is largely independent of the detailed knowledge of how our organism works. Namely, with current studies of diseases (instead of robust health) and by favouring applied (instead of fundamental) research, we risk waiting a long time for such useful knowledge**. Particularly attractive is the principle that same protein protection by molecular chaperoning applies to prevention and healing of diseases.

---

*One such molecule, isolated from a UV radiation-resistant bacterium, shows the properties of an antioxidant chemical chaperone: it interacts with misfolded proteins and protects them from UV-induced oxidative carbonylation.

** French Nobel Prize laureate, the physicist Jean Perrin, is quoted for saying: if medical doctors were given the task to find a way to look inside living organisms, we still would not have X-rays.